

Questions 1–5: 5 points each, questions 6–10: 15 points each.

1. Which one of the following is the linearization of $f(x, y) = \ln(x^2 + y^2)$ at the point $(1, -1)$?

- (a) $L(x, y) = -x + y + \ln 2 - 2$ (c) $L(x, y) = x - y + \ln 2$
 (b) $L(x, y) = x - y + \ln 2 - 2$ (d) $L(x, y) = -x + y + \ln 2$

2. The limit of $f(x, y) = \tan^{-1} \left(\frac{|x| + |y|}{x^2 + y^2} \right)$ as $(x, y) \rightarrow (0, 0)$ is

- (a) 0 (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{2}$ (d) Undefined

3. If $\mathbf{F}(x, y, z) = \nabla g(x, y, z)$, then $\text{div}(\mathbf{F}) =$

- (a) 0 (b) $\text{curl}(\mathbf{F})$ (c) $g_{xx} + g_{yy} + g_{zz}$ (d) $g_{xx} \mathbf{i} + g_{yy} \mathbf{j} + g_{zz} \mathbf{k}$

4. Let C be any smooth curve from the point $A(0, 2, 10)$ to the point $B(1, 1, \pi)$. If $f(x, y, z) = y^2 e^{x \sin z}$, then $\int_C \nabla f \cdot d\mathbf{r} =$

- (a) 3 (b) -3 (c) 0 (d) $4 - e^{-1}$

5. Let $\mathbf{F}(x, y, z)$ be a vector field with component functions that have continuous partial derivatives over $\mathbb{R}^3$. If $\mathbf{F}$ is a conservative field, then

- (a) $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path. (c) $\nabla \times \mathbf{F} = \mathbf{0}$.
 (b) $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every piecewise-smooth closed curve. (d) All of the above.

6. Find the minimum value of $f(x, y) = y^2 - x^2 - 2x$ subject to the constraint $x^2 + y^2 = 1$.

7. Evaluate $\iint_S \sqrt{x^2 + y^2} \, dS$, where S is the helicoid given by the parametrization

$$\mathbf{r}(u, v) = \langle u \cos v, u \sin v, v \rangle, \quad 0 \leq u \leq 1, \quad 0 \leq v \leq 2\pi.$$

8. (a) Evaluate $\int_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$, where C is the unit circle centered at $(0, 0)$, oriented counterclockwise.
 (b) Explain why Green's Theorem cannot be applied to evaluate this integral.

9. (a) State the Divergence Theorem.

- (b) Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{F}$ is the vector field

$$\mathbf{F}(x, y, z) = \left[x^3 + \arctan(zy^2) \right] \mathbf{i} + \left[x e^{\sin z} - yx^2 \right] \mathbf{j} + \left[2zy^2 - \cos(e^x) \right] \mathbf{k},$$

and S is the positively oriented surface boundary of the solid enclosed by the inverted cone $z = 2 - \sqrt{x^2 + y^2}$ and the xy -plane.

10. Let C be the curve of intersection of the paraboloid $z = x^2 + y^2$ and the plane $x + y + z = 1$, with a counterclockwise orientation when viewed from above. Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}$ is the vector field $\mathbf{F}(x, y, z) = 2xz \mathbf{i} + 3z^2 \mathbf{j} + (x^2 + 6yz) \mathbf{k}$.

SOLUTIONS

1. The correct choice is (b).

$$f_x = 2x/(x^2 + y^2) \text{ and } f_y = 2y/(x^2 + y^2) \implies f(1, -1) = \ln 2, f_x(1, -1) = 1, \text{ and } f_y(1, -1) = -1.$$

$$\text{These yield } L(x, y) = \ln 2 + 1(x - 1) + (-1)(y + 1) = x - y + \ln 2 - 2.$$

2. The correct choice is (b).

$$\text{Using polar coordinates we have } \lim_{(x,y) \rightarrow (0,0)} \frac{|x| + |y|}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \frac{r(|\cos \theta| + |\sin \theta|)}{r^2} = \lim_{r \rightarrow 0^+} \frac{|\cos \theta| + |\sin \theta|}{r}.$$

Note that $|\cos \theta| + |\sin \theta| \geq 1$ for all θ . Hence $\frac{|\cos \theta| + |\sin \theta|}{r} \geq \frac{1}{r} \rightarrow \infty$ as $r \rightarrow 0^+$, which yields

$$\lim_{(x,y) \rightarrow (0,0)} \frac{|x| + |y|}{x^2 + y^2} = \infty. \text{ In turn we obtain } \lim_{(x,y) \rightarrow (0,0)} \arctan\left(\frac{|x| + |y|}{x^2 + y^2}\right) = \frac{\pi}{2}.$$

3. The correct choice is (c).

$$\operatorname{div} \mathbf{F} = \operatorname{div}(\nabla g) = \operatorname{div}(g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}) = g_{xx} + g_{yy} + g_{zz}.$$

4. The correct choice is (b).

$$\text{By the Fundamental Theorem for Line Integrals, } \int_C \nabla f \cdot d\mathbf{r} = f(1, 1, \pi) - f(0, 2, 10) = 1 - 4 = -3.$$

5. The correct choice is (d).

6. Let $g(x, y) = x^2 + y^2$. By the method of Lagrange multipliers, the minimum occurs at a point on the circle $g(x, y) = 1$ where $\nabla f = \lambda \nabla g$. That is, a solution of the system

$$\begin{cases} -2x - 2 = \lambda \cdot 2x \\ 2y = \lambda \cdot 2y \\ x^2 + y^2 = 1. \end{cases}$$

- The second equation gives $y(1 - \lambda) = 0$ which holds if and only if $y = 0$ or $\lambda = 1$.
- If $y = 0$, the last equation gives $x = \pm 1$.
- If $\lambda = 1$, the first equation becomes $-2x - 2 = 2x$ which holds if and only if $x = -\frac{1}{2}$. Now the last equation yields $y = \pm \frac{\sqrt{3}}{2}$.

Hence the desired minimum value occurs at one of the points $(\pm 1, 0)$ and $(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2})$.

Now since $f(-1, 0) = 1$, $f(1, 0) = -3$, and $f(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2}) = \frac{3}{2}$, we conclude that the minimum value of $f(x, y)$ on the circle $x^2 + y^2 = 1$ is $f(1, 0) = -3$.

7. We have

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos v & \sin v & 0 \\ -u \sin v & u \cos v & 1 \end{vmatrix} = \langle \sin v, -\cos v, u \rangle \implies |\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{1 + u^2}.$$

Therefore,

$$\begin{aligned} \iint_S \sqrt{x^2 + y^2} \, dS &= \int_0^{2\pi} \int_0^1 \sqrt{u^2 \cos^2 v + u^2 \sin^2 v} \sqrt{1 + u^2} \, du \, dv \\ &= \int_0^{2\pi} \int_0^1 u \sqrt{1 + u^2} \, du \, dv = 2\pi \int_0^1 u \sqrt{1 + u^2} \, du = \frac{2\pi}{3} (1 + u^2)^{3/2} \Big|_0^1 = \frac{2\pi}{3} (2\sqrt{2} - 1). \end{aligned}$$

8. (a) The circle C may be parametrized as $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j}$ where $0 \leq t \leq 2\pi$. This gives

$$\int_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = \int_0^{2\pi} \left[\frac{-\sin t}{1}(-\sin t) + \frac{\cos t}{1}(\cos t) \right] dt = \int_0^{2\pi} dt = 2\pi.$$

- (b) Let D be the unit disk $x^2 + y^2 \leq 1$ enclosed by C . Green's Theorem cannot be applied since the functions $P(x, y) = -y/(x^2 + y^2)$ and $Q(x, y) = x/(x^2 + y^2)$ are undefined at $(0, 0) \in D$.

9. (a) Let E be a simple solid region and let S be the boundary surface of E with outward orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E . Then
$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \operatorname{div} \mathbf{F} dV.$$

- (b) The solid E enclosed by S is described in cylindrical coordinates as

$$E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, 0 \leq z \leq 2 - r\}.$$

It is easy to see that all conditions of the Divergence Theorem are satisfied and hence

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_E \operatorname{div} \mathbf{F} dV = \iiint_E (3x^2 - x^2 + 2y^2) dV \\ &= \int_0^{2\pi} \int_0^2 \int_0^{2-r} 2r^2 r dz dr d\theta = 2\pi \int_0^2 2r^3(2-r) dr = 2\pi \left[r^4 - \frac{2}{5}r^5 \right]_0^2 = \frac{32\pi}{5}. \end{aligned}$$

10. Let S be the part of the plane $x + y + z = 1$ that is bounded by the curve C . Since

$$\operatorname{curl} \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xz & 3z^2 & x^2 + 6yz \end{vmatrix} = \langle 6z - 6z, 2x - 2x, 0 - 0 \rangle = \langle 0, 0, 0 \rangle,$$

by Stokes' Theorem we get
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} dS = \iint_S 0 dS = 0.$$